OtterFind - Requirements

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Team: p3_s1_group_07 (Group 7) · **Repo:** https://github.com/CST438OtterFind/OtterFind · **Last updated:** 11/2/26

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1. Users

- **Primary user:** a CSUMB student or staff member who has lost something on campus (a pet, a laptop, a notebook, keys) and wants people who walk past it to be able to tell them exactly where they saw it.
- **Secondary user:** A CSUMB student or staff member that finds a lost item can create a report for it with any additional details.
- **Tertiary user:** An admin that removes any abusive or fake posts/messages.

2. Problem

Two sentences: what the user struggles with today, and why existing options fall short.

With the bustle of school life, it's not unheard of to leave something behind in a classroom by accident. Currently, there are very limited options for a "lost and found" system, since not everything valuable on your person is able to be tracked.

3. Success measures

Three measurable signals that would show the product works. Each one needs a number and a way to check it.

#	Measure	Target	How we check it
S1	A new user signs in and posts a lost item with its building and a map pin	Under 1 minute 30 sec	Timed with a visiting student during the onboarding check-in
S2	A user is able to click on a lost item and see its details as well as its location on the map.	Under 10 seconds	When clicking on an item, the screen transitions/adds item details to the view without crashing the app.
S3	A user is able to communicate and send a message to someone who has found an item	Under 1 minute	When selecting a report for a lost item, a message function shows up and allows a user to communicate successfully.

4. Scope: MVP, vertical slice, and not building

MVP is the smallest set of stories that makes *this* product useful. It always includes the account baseline (sign in with GitHub and Google, update profile, delete account) **plus** the stories specific to your product.

- **MVP stories:** `PRD-1, PRD-2, PRD-3, PRD-4, PRD-5, PRD-6, PRD-7` (listed in section 5), plus the account baseline: sign in with GitHub and Google, update profile, delete account.
- **Vertical slice:** PRD-4: Post a lost item with where it was last seen, then a second user finds it and adds a sighting (PRD-7) the owner can see, on web and Android. This is what the Playwright test walks and what the demo opens with.
- **Not building (this term):**
  - Anything beyond the CSUMB campus.
  - An iOS app (we ship web and Android only).
  - Rewards or payments for returning an item.
  - Ownership verification or claim checks for found items.

5. User stories and acceptance criteria

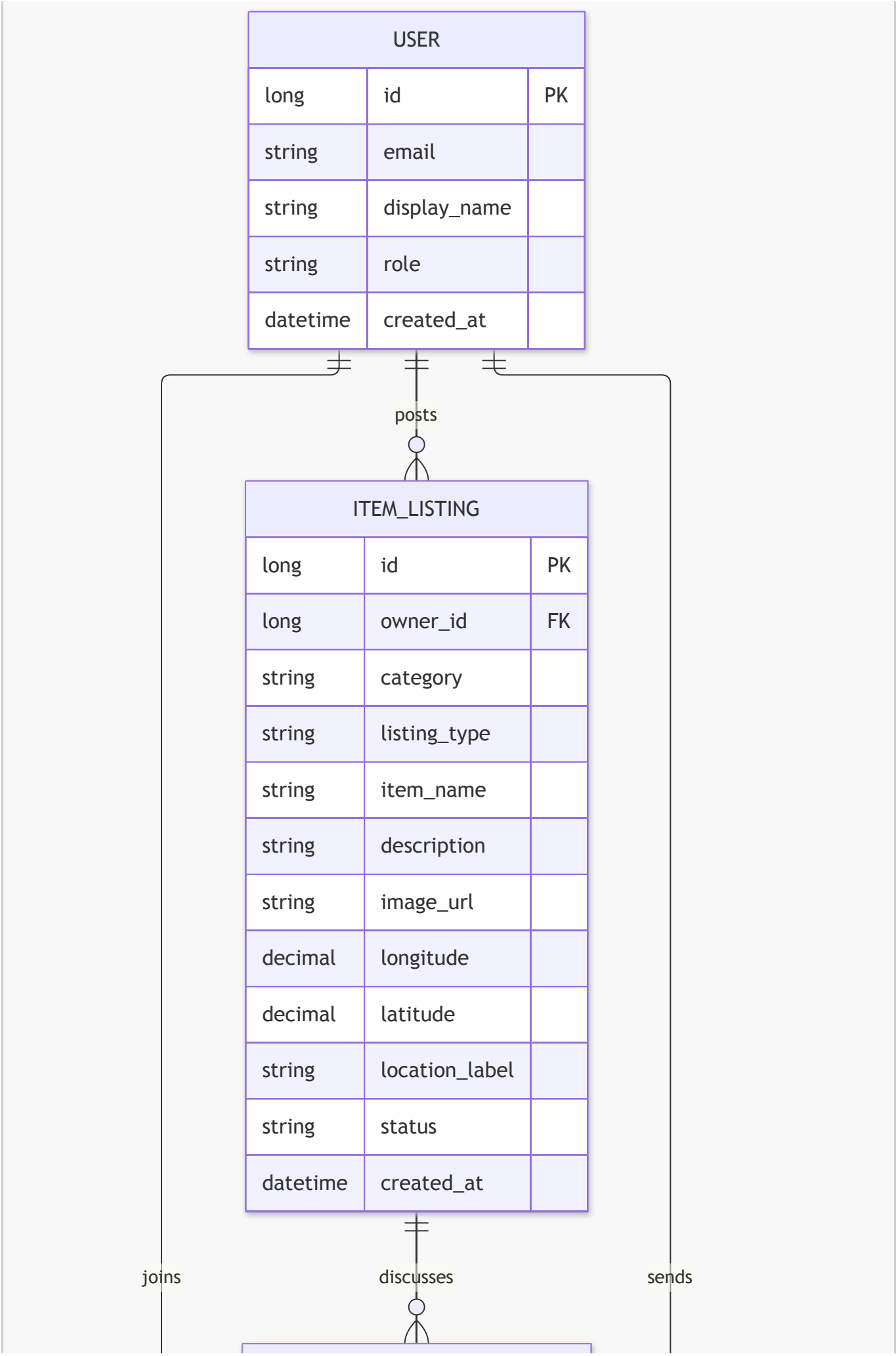
One row per MVP story. Each GitHub issue links to its story ID (`PRD-3`) and each story has at least one acceptance criterion in **Given / When / Then** form that someone outside the team could test.

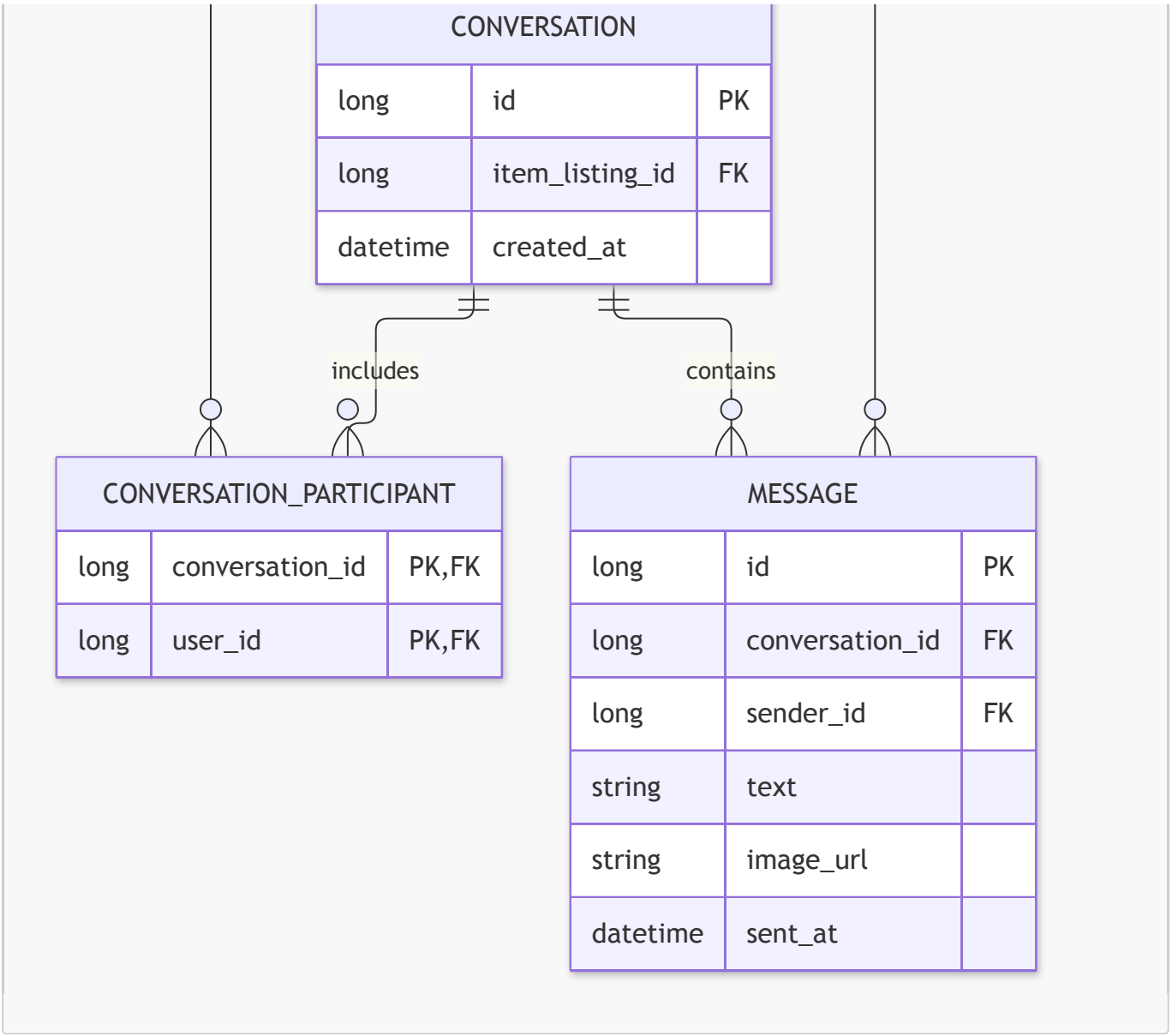
ID	Story	Acceptance criteria
PRD-1	As a user, I want to sign in with GitHub or Google, update my profile, and delete my account so that I control my identity on OtterFind.	**Given** I am signed out, **when** I choose "Sign in with Google" or "Sign in with GitHub" and approve, **then** I land on the feed with my name shown. **Given** I am signed in, **when** I change my display name and save, **then** the new name shows on my profile and my posts. **Given** I am signed in, **when** I delete my account and confirm, **then** I am signed out and my profile and posts are removed.
PRD-2	As a staff member who found multiple lost items, I want to see a list of reported lost items so I can try and pinpoint who they belong to.	**Given** `I have an account and am signed in`, **when** `I pull up the map or list of lost items`, **then** `I can then find more details on the lost items.`
PRD-3	As someone who found an item with an unknown owner, I want to be able to report a missing item to try and find the owner.	**Given** `I have an account and found a missing item`, **when** `I fill out a Report with details`, **then** `the lost item is posted for other users to see.`
PRD-4	As someone who lost something, I want to post it with where I last saw it so that people know what and where to look.	**Given** I am signed in, **when** I post "Grey tabby cat" as Lost, category Pet, building "Library", floor 2, a pin on the map and a last-seen time, **then** it appears at the top of the feed with its building and pin. **Given** a title under 4 characters or no building, **when** I submit, **then** I see which field is wrong and nothing is posted.

ID	Story	Acceptance criteria
PRD-5	As someone who wants to help the person who lost the item, I want to be able to find where the item was last seen and go to that place on the map.	**Given** I am signed in and a lost item "Grey tabby cat" was posted with building "Library" and a map pin, **when** I open the item and tap "Show on map", **then** the map opens centered on that pin with the building and last-seen time shown. **Given** an item has no map pin, **when** I open it, **then** I see its building only and no "Show on map" button.
PRD-6	As someone who lost or found an item, I want to message the other person in the app so that we can arrange to return it without sharing personal contact info.	**Given** I am signed in and viewing someone else's post, **when** I tap "Message" and send "I think I saw your cat by the library", **then** the conversation appears in both users' inboxes with the post linked. **Given** I am the post owner, **when** I reply, **then** the other user sees my reply in the same conversation. **Given** I am viewing my own post, **then** no "Message" button is shown.
PRD-7	As someone who spotted a lost item, I want to add a sighting with where and when I saw it so that the owner knows where it is now.	**Given** I am signed in and viewing "Grey tabby cat", **when** I add a sighting with building "Student Center", a map pin and the time, **then** the sighting appears on the item, newest first, and the owner sees it on their post. **Given** I leave the building empty, **when** I submit, **then** I see which field is wrong and nothing is added.

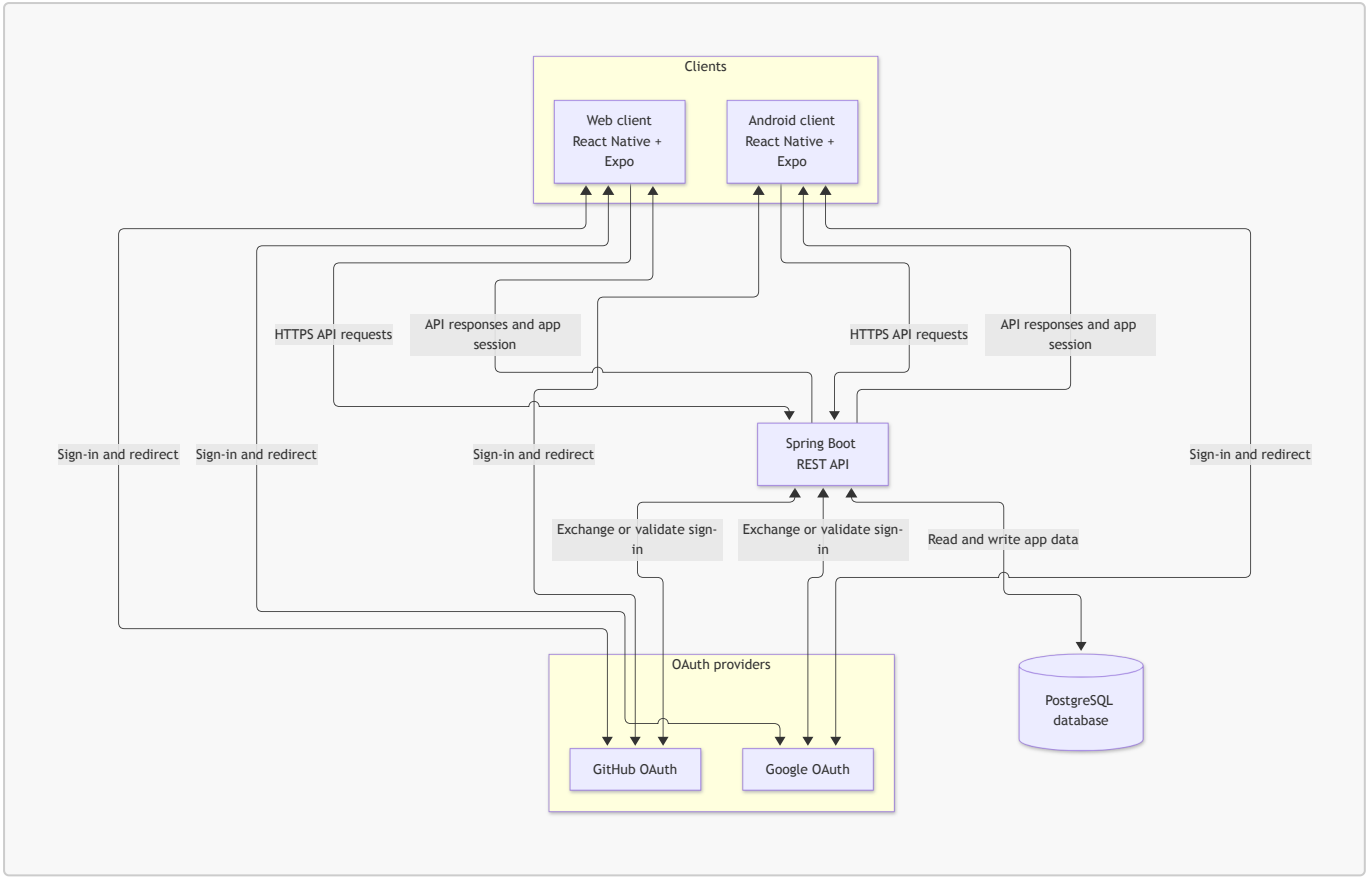
6. Technical Plan

Lost and Found ERD


USER
long id PK
string email
string display_name
string role
datetime created_at
posts
ITEM_LISTING
long id PK
long owner_id FK
string category
string listing_type
string item_name
string description
string image_url
decimal longitude
decimal latitude
string location_label
string status
datetime created_at
joins
discusses
sends



Architecture Sketch



API Routes for the vertical slice

Method	Route	Purpose
POST	/api/auth/{provider}/exchange	Complete sign-in with Google or GitHub and return the app session
GET	/api/users/me	Return the signed-in user
POST	/api/uploads	Upload an item or message photo and return its URL
GET	/api/items?listingType=&lat=&lng=&radius=	Browse listings, optionally filtered by type and location
POST	/api/items	Create a lost or found listing
GET	/api/items/{itemId}	View a listing’s details
POST	/api/items/{itemId}/conversations	Start a conversation about the listing
GET	/api/conversations	List the signed-in user’s conversations
GET	/api/conversations/{conversationId}/messages	Load messages in a conversation
POST	/api/conversations/{conversationId}/messages	Send a message, optionally with an uploaded image URL

Where is everything hosted?

| Website: Hosted on Render |

| Our API: Springboot Docker Container running on render |

| Android App: Expo application using React Native |

| Database: Supabase |

Change log

Date	PR	What changed and why
10/29	1	Brandon - added PRD, success target,
11/1	2	Daniel - updated PRD
11/2	4	Daniel - updated PRD
11/2	5	Tony - updated PRD
11/2	6	Rian - updated PRD, added ERD, Mermaid Chart